Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A heat engine operates between **500 K** and **300 K** and absorbs **700 J**. **Find maximum ideal work** and **explain why actual work** is usually lower.

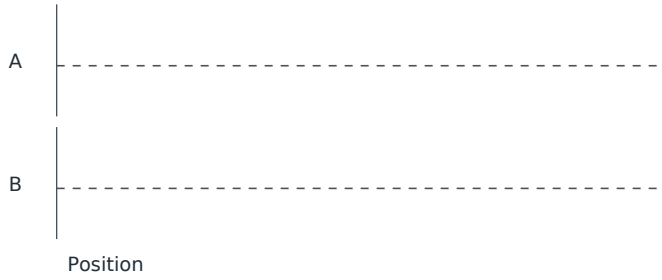
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: v = f λ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength **3 m** and frequency **2 Hz** travels at v=f λ=**6 m/s**. At the same speed, frequency **3 Hz** would require wavelength **2 m**.

Your turn: Draw both waves over the same **6 m** length and **label their frequencies**.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

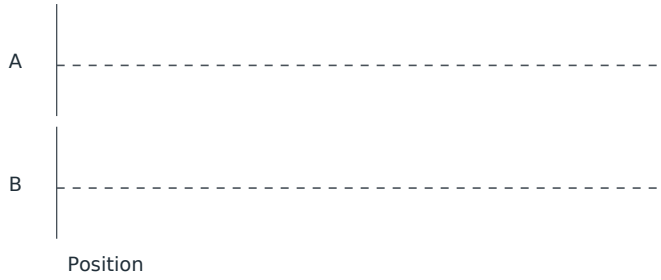
Two waves travel in the same ideal medium at the same speed. One has twice the frequency. **Compare wavelengths**.



Marzano check - circle: ① ② ③

04 Apply the physics

A speaker produces a **425 Hz** tone in air where sound speed is **340 m/s**.
1. **Calculate its wavelength**.
2. **Find the frequency** that would produce a **0.50 m** wavelength in the same air.
3. **Sketch the relative crest spacings**.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student writes v=f/λ. **Check units** and **explain why** that is not a speed.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An object at **360 K** sits in a vacuum facing surroundings at **240 K**. How can it cool, and are absolute-zero surroundings required?
Label temperatures, samples and relevant energy transfers.

Sample A

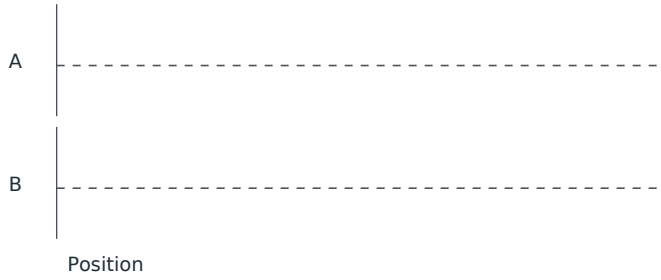
Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A wave travels **12 m** in **0.40 s** and has frequency **50 Hz**. **Find speed** and wavelength **0**.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A wave enters a medium where speed changes from **12** to **8 m/s**. Frequency stays **4 Hz**. **Find both wavelengths** and **explain why frequency** does not reset at the boundary.



Marzano check - circle: 1 2 3